Dijkstra's Algorithm:

(I) Initialize - Single - Source (G, S).

1. For each vertex $v \in V(G)$
2. $\quad d(v) \leftarrow \infty$
3. $\quad \pi(v) \leftarrow$ Nil
4. $d[S] \leftarrow 0$

(II) Relax(u, v, w)

1. If $d[u] + w(u,v) < d(v)$
2. then $d[v] \leftarrow d[u] + w(u,v)$
3. $\pi[v] \leftarrow u$

(III) Dijkstra(G, w, s)

1 Initialize - Single- Source(G, s)
2. $S \leftarrow \phi$
3. $Q \leftarrow V(G)$

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4. while $Q \neq \phi$

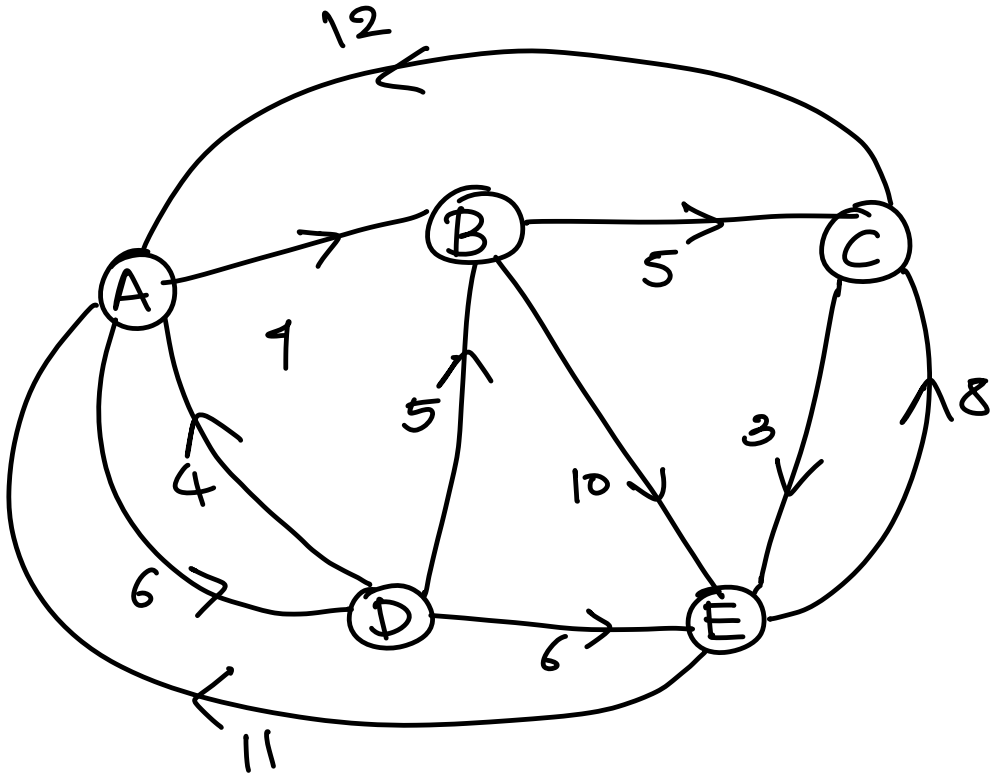
5. $u \leftarrow$ Extract-Min(Q)

6. $S \leftarrow S \cup \{u\}$

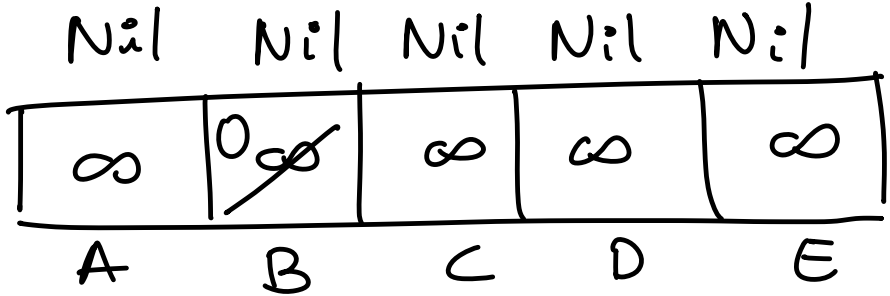
7. For each vertex $v \in V(G)$

8. do Relax (u, v, w).

Eg:



Initialization:



Start vertex = B

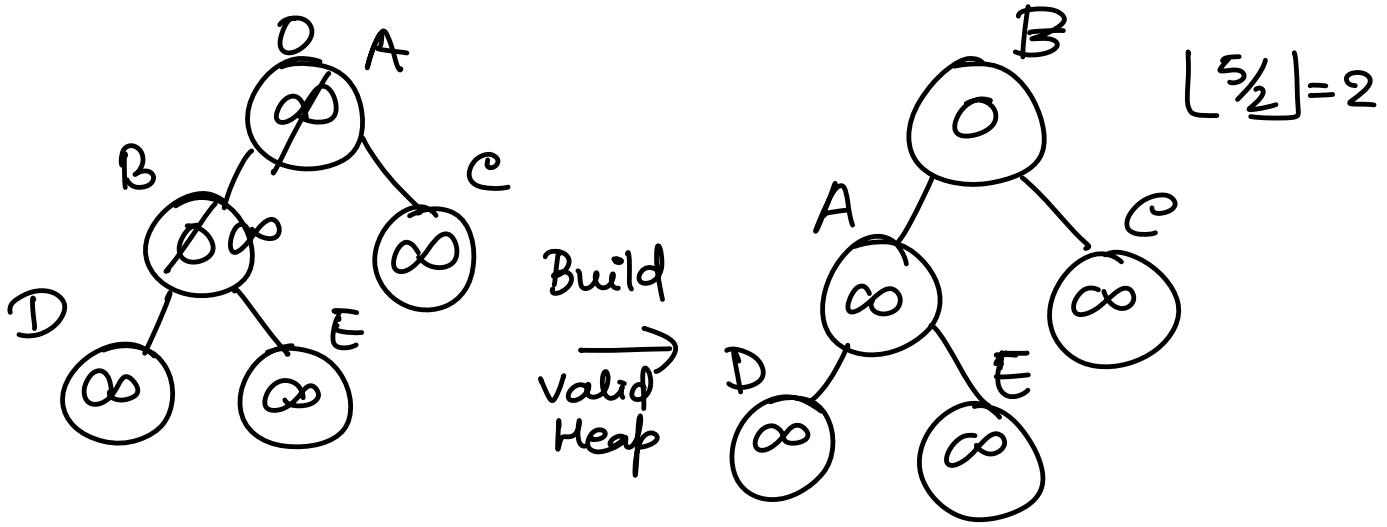
	A	C	D	E
B (Starting vertex)	∞ (No edge)	Min 5 (B)	∞	10 (B)
C (wt = 5)	$17<\infty$ (5+12) 17 (C)	✓	∞	$8<10$ (5+3) 8 Min (C)
E (wt = 8)	$8+11=19 \not< 17$ 17 Min (C)	✓	∞	✓
A (wt = 17)	✓	✓	$23<\infty$ (17+6) 23 (A)	✓
	✓	✓	✓	✓

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	A	C	D	E
B	17 Through C	5 Through B	23 Through A	8 Through C

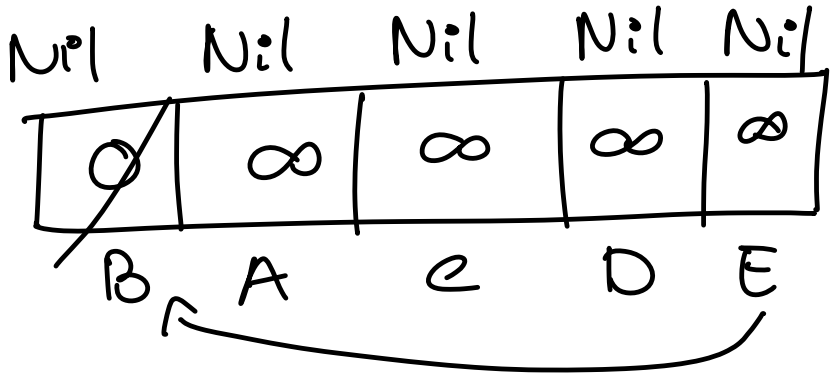
$S = \phi$ $\quad Q = V(G)$.

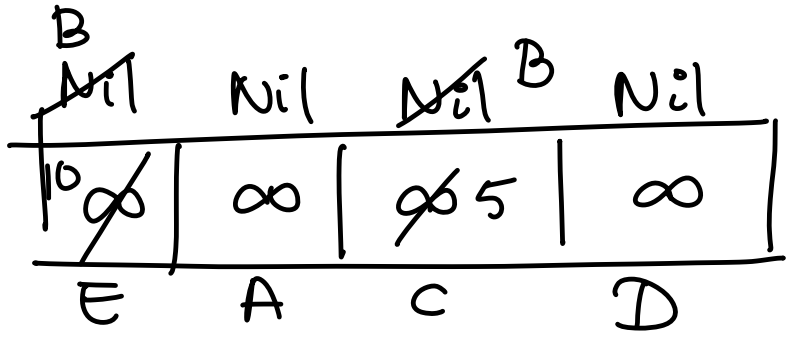
$Q \neq \phi$.



$u \leftarrow B$

$S = \{\} \cup \{B\} = \{B\}$.





$V(G) = \{E, A, C, D\}$ $\quad u = B.$

Relax (u, v, w)

$=$ Relax (B, v, w) $\quad v \in V(G)$

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$d[u] = d[B] = 0$.

$V = E$ $\quad \omega(B, E) < d[E]$

$10 < \infty$ (update)

A $\quad \omega(B, A) < d(A)$

$\infty < \infty$ (No update)

C $\quad \omega(B, C) < d(C)$

$5 < \infty$ (update)

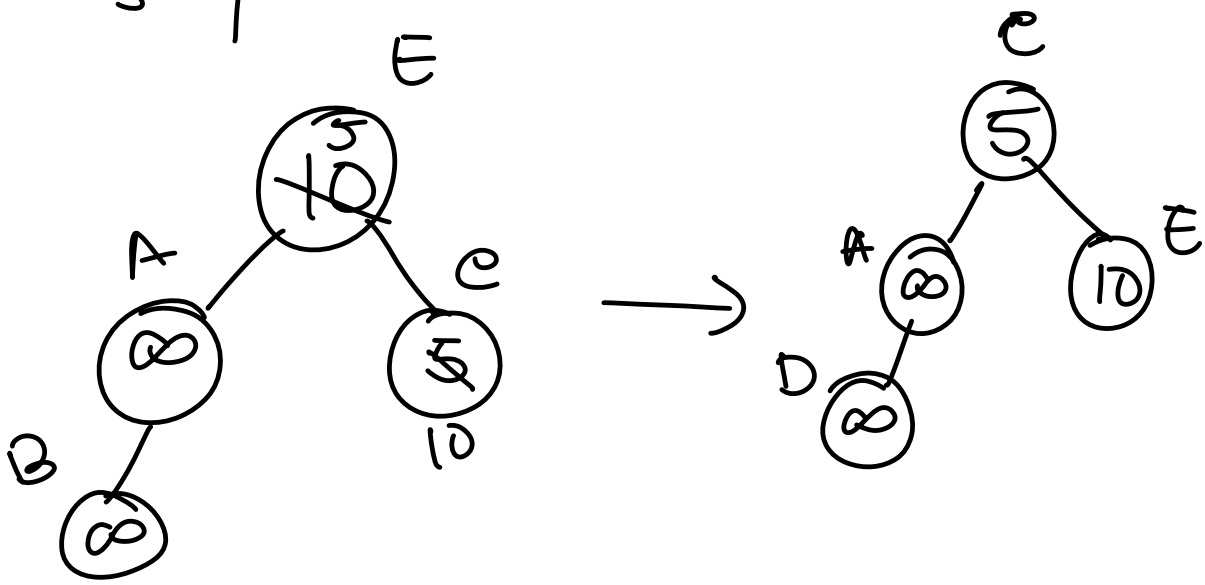
D $\quad \omega(B, D) < d(D)$

$\infty < \infty$ (No update)

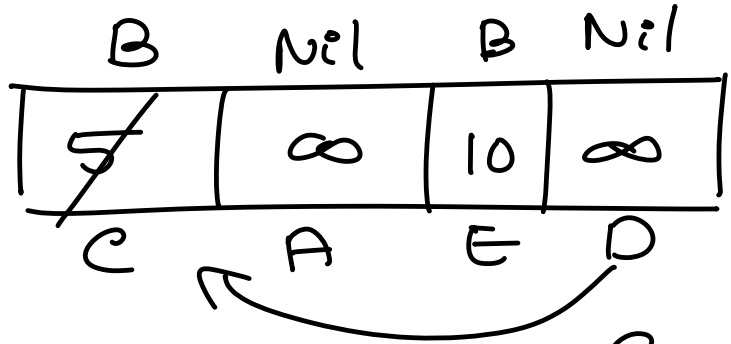
Q

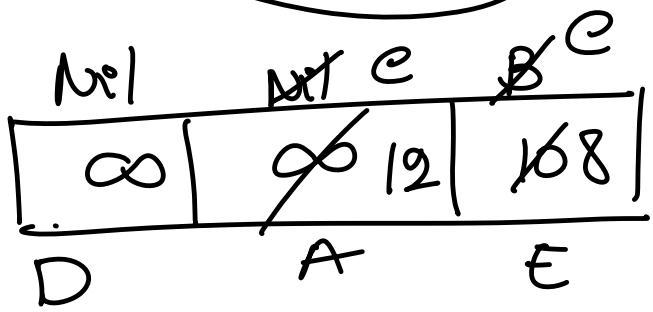
B	Nil	B	Nil
10	∞	5	∞
E	A	C	D

$Q = \phi$



$\lfloor 4/2 \rfloor = 2$





$u \leftarrow C$

$S = \{B, C\}$

$V(G) = \{D, A, E\}$

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Relax (C, v, ω) $d(C) = 5$ $\forall v \in V(G)$

$V = D$ $\quad d(C) + \omega(C, D) < d(C)$

$5 + \infty \not< \infty$ (No update)

$V = A$ $\quad d(C) + \omega(C, A) < d(A)$

$5 + 12 < \infty$ (Update)

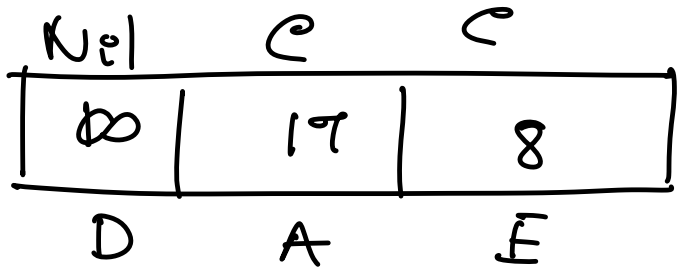
$= 17$

$V = E$ $\quad d(C) + \omega(C, E) < d(E)$

$5 + 3 < 10$ (update)

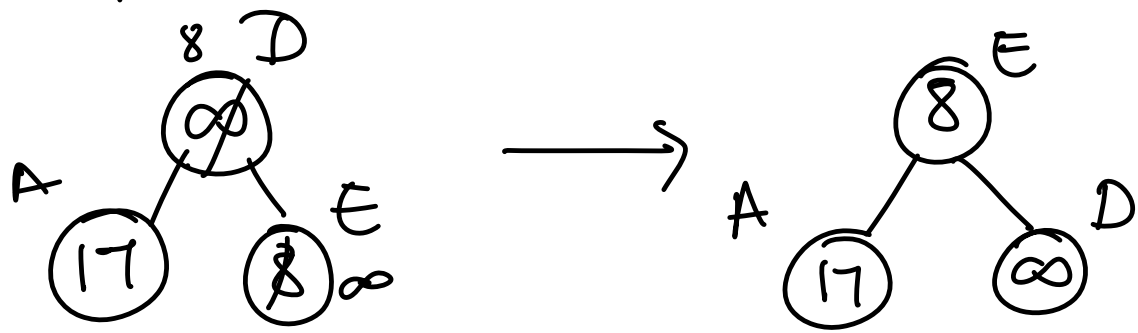
$= 8$

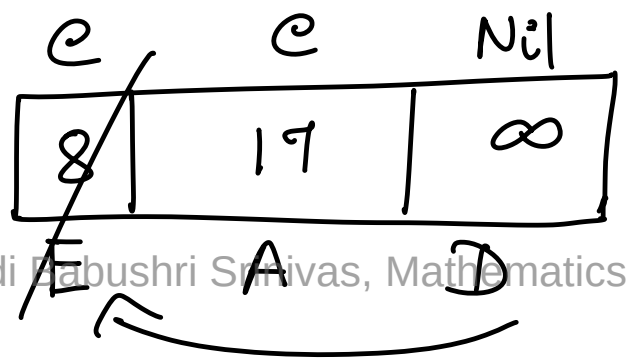
Q:



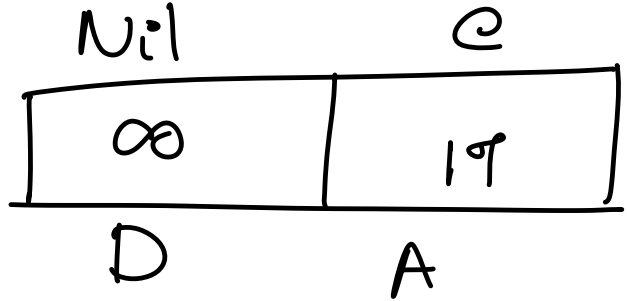
$Q \neq \phi$

$\lfloor 3/2 \rfloor = 1$





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$u \leftarrow E$

$S = \{B, C, E\}$

$V = \{D, A\}$ $\quad d(E) = 8$

$V = D$ $\quad d(E) + \omega(E, D) < d(D)$

$8 + \infty \not< \infty$ (No update)

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$V = A$ $\quad d(E) + \omega(E, A) < d(A)$

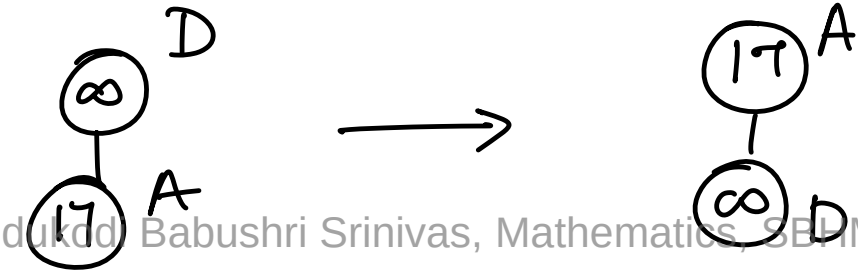
$8 + 11 \not< 17$ (No update)

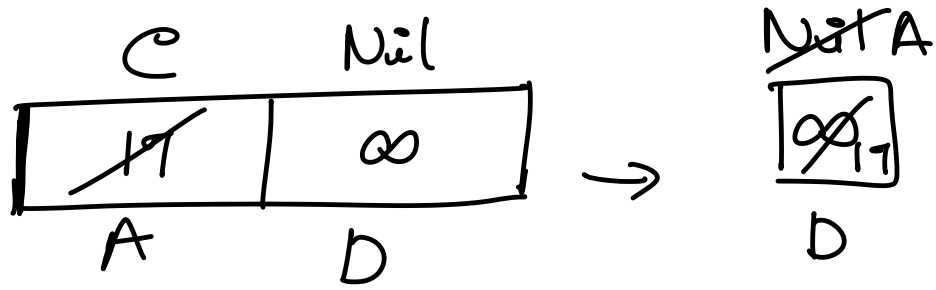
$= 19$

Nil	C
∞	19
D	A

(Remains same)

$Q = \phi$





$u \leftarrow A$

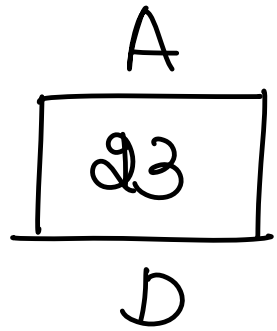
$S = \{B, C, E, A\}$

$V(G) = \{D\}$ $\quad d(A) = 19$

$V = D$ $\quad d(A) + \omega(A, D) < d(D)$

$19 + 6 < \infty$ (update)

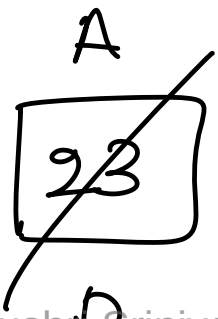
$= 23$



$Q \neq \phi$



Valid Heap



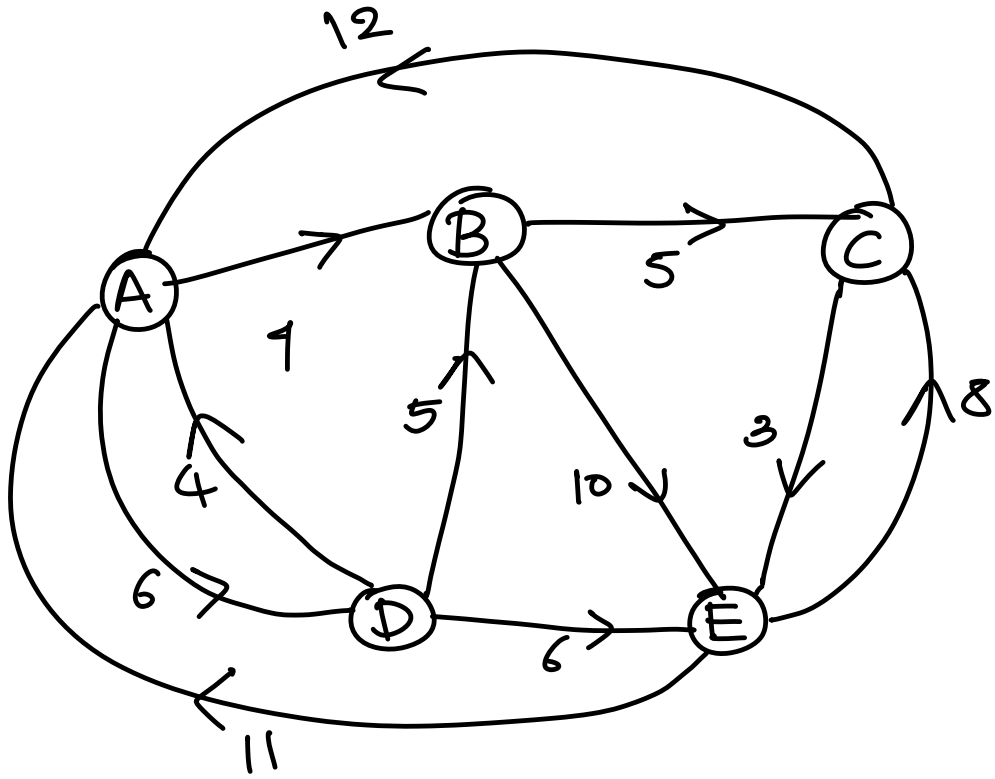
$u \leftarrow D \qquad d(D) = 23$

$V(G) = \{\ \}$

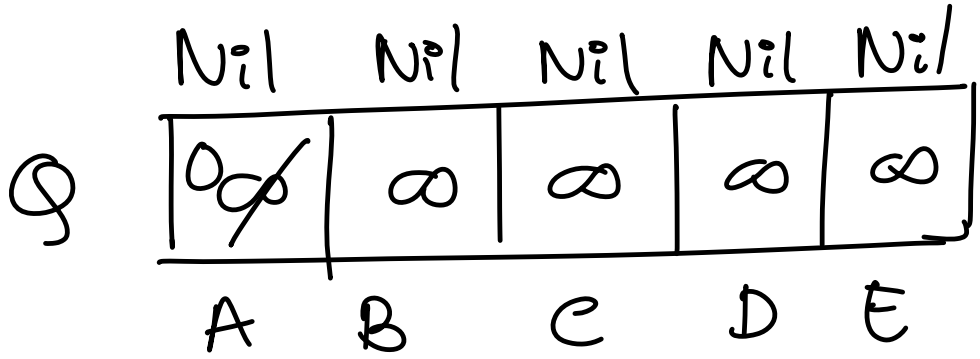
$Q = \phi$

(Stops).

B	C	E	A	D
Distance	5	8	17	23
Through	B	C	C	A

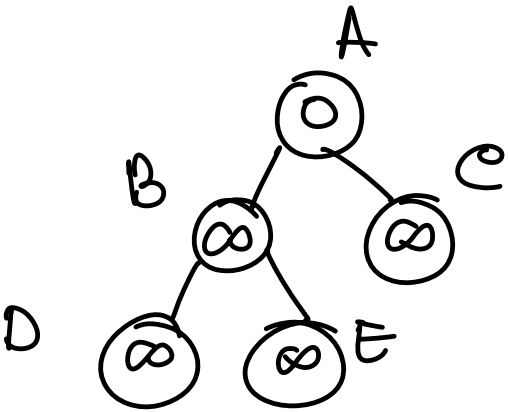


Starting Vertex = A

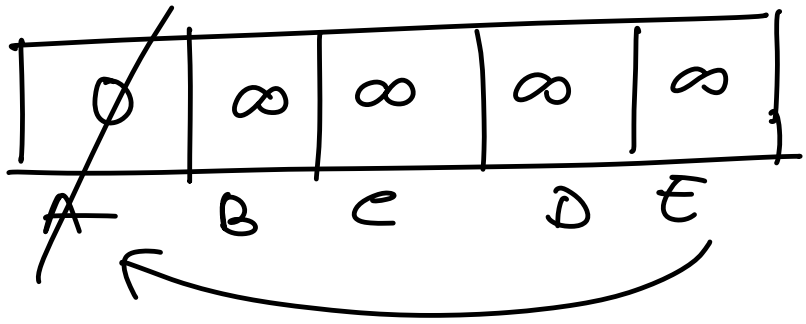


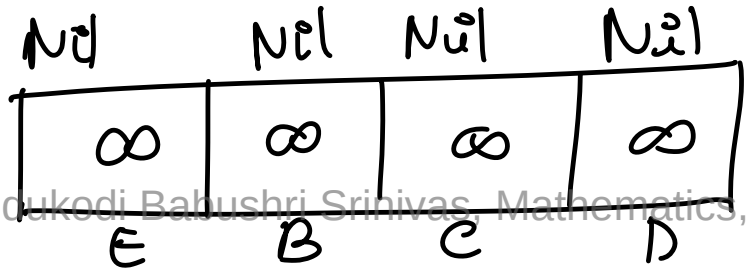
$S = \{\ \}$

$Q \neq \phi$



(Valid Heap)





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$u \leftarrow A \qquad d(A) = 0$

$V(G) = \{E, B, C, D\}$

$V = E \qquad \omega(A,E) < d(E)$

$\infty \not< \infty$

$V = B \qquad \omega(A,B) < d(B)$ (update)

$7 < \infty$

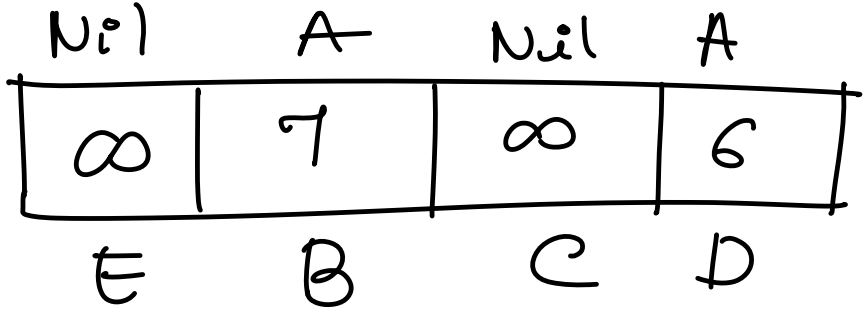
$V = C \qquad \omega(A,C) < d(C)$

$\infty \not< \infty$

$V = D \qquad \omega(A,D) < d(D)$ (update)

$6 < \infty$

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Nil
A
Nil
A
∞
7
∞
6
E
B
C
D

Permutation Puzzles:

16- Puzzle

1	2	3	4
5	6	7	8
9	10	11	12
13	15	14	

Given State



1	2	3	4
5	6	7	8
9	10	11	12
13	14	15	

Solved State.